

AUTOMATA

VIRAL POTENTIAL ANALYZER

Lecturer : Mr. Lykheang Peng
Student : Hak Chandyna

Content

Content

1. Problem Statement
2. Project Objective
3. Source
4. Expectation
5. System Overview
6. Dataset
7. Methodology
8. Automata Design
9. Results
10. Conclusion

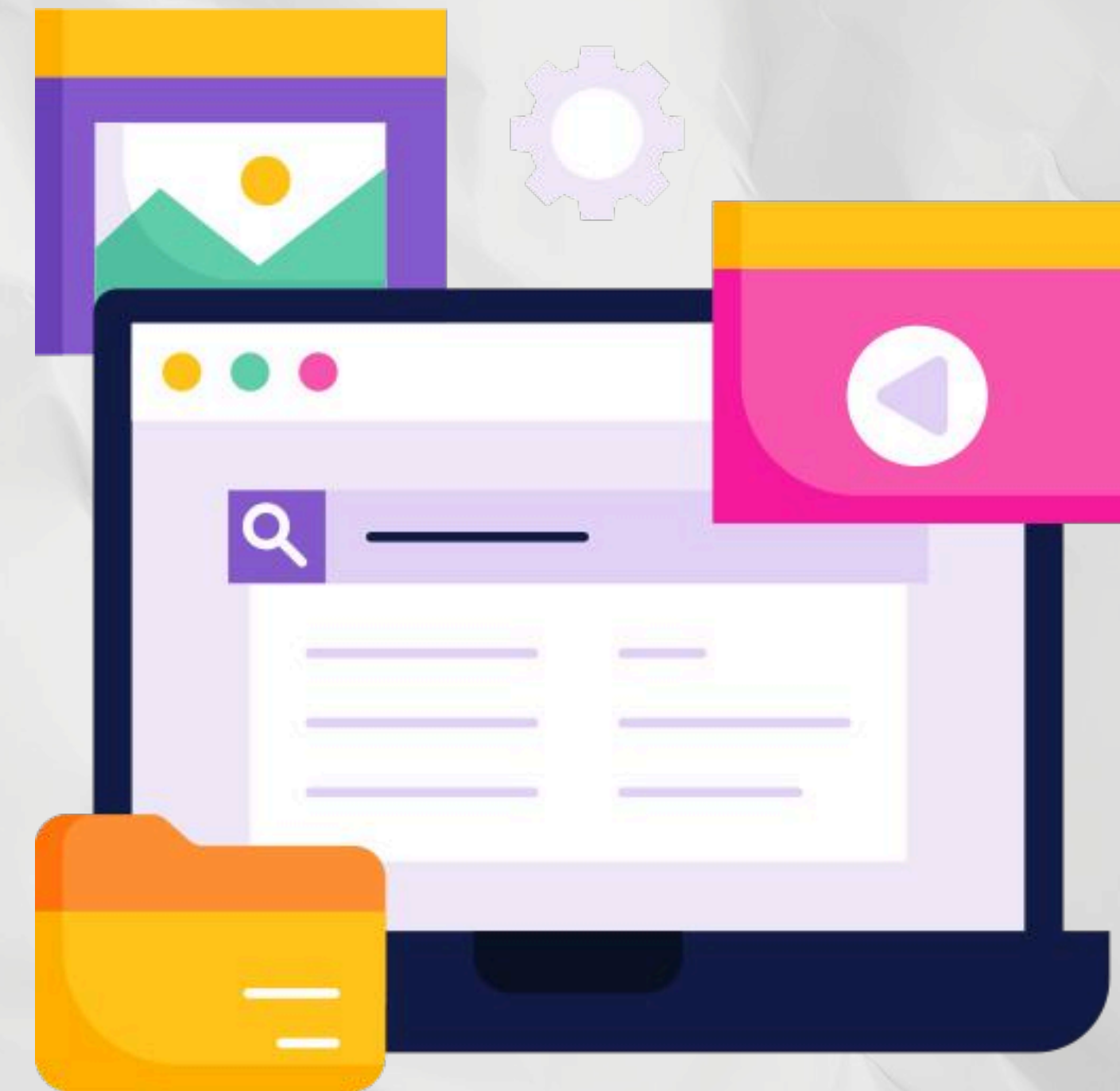
Problem Statement

Problem Statement

Content creators often face challenges in understanding the various factors that impact the performance of their content.

This includes selecting the appropriate hashtags, audio, timing, and video length, as making the wrong choices in these areas can lead to decreased engagement levels.

Therefore, there is a clear need for a robust system that can analyze content comprehensively before it is published.



Project Objective

IDT

Project Objective

- Analyze content features that influence engagement
- Calculate a content performance score
- Classify viral potential using automata
- Provide improvement recommendations

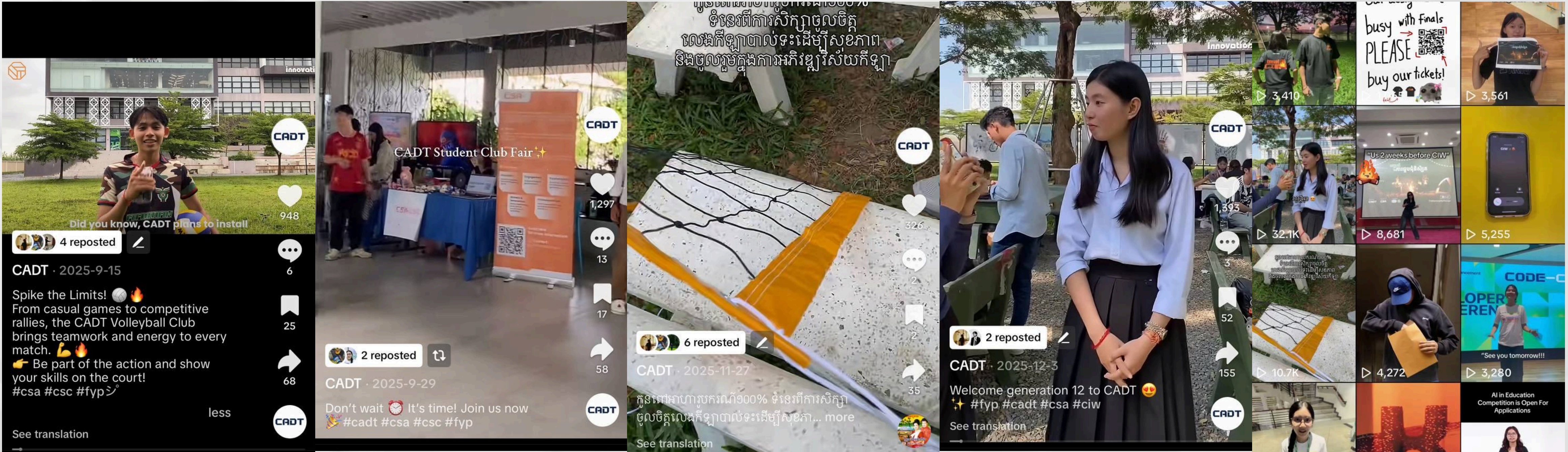


Source?

TRUST ME BRO

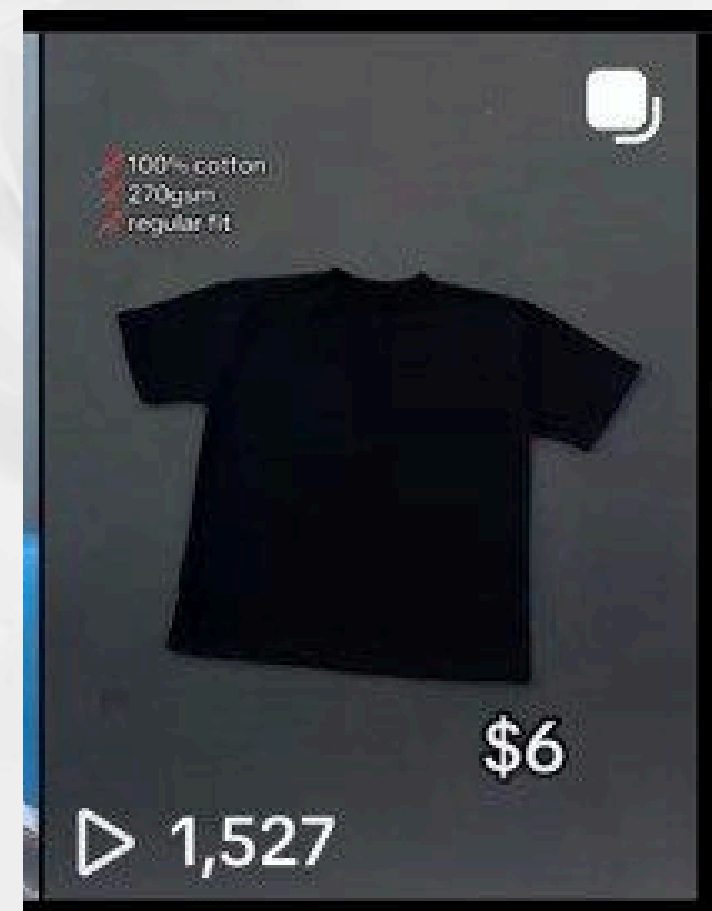


Source

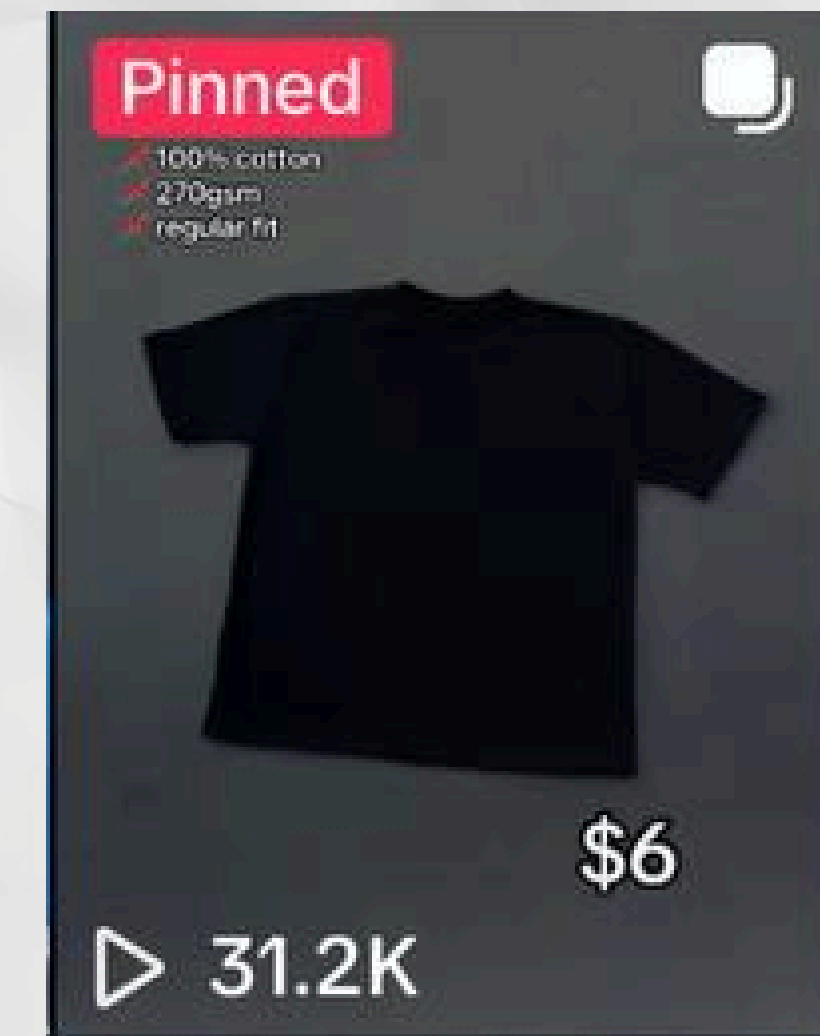
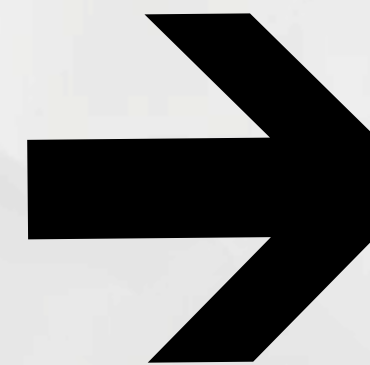


Expectation

Expectation



IDT



System Overview

IDT

System Overview

Overview of workflow



Dataset

IDT

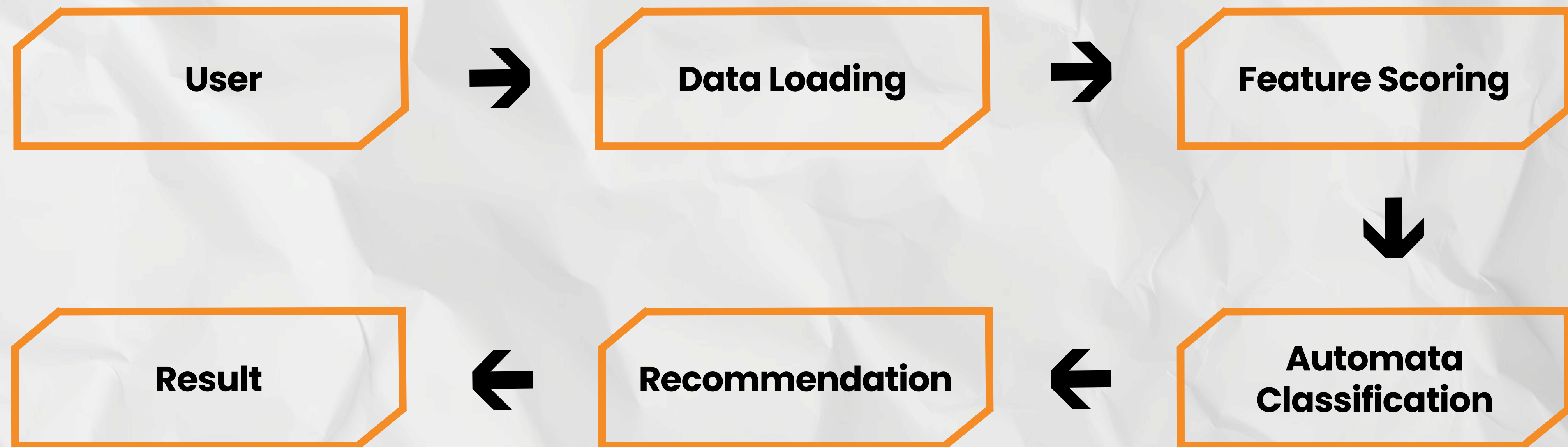
Dataset

	Feature	Description
Content	content_type	Type of content
	category	Content category
	points	Score
Hashtag	hashtag	Hashtag name
	trend_level	Popularity
	points	Score
	category	Relevance
	recommended_for	Recommendation
Audio	audio_name	Audio title
	trend_level	Popularity
	points	Score
	recommended_for	Recommendation

	Feature	Description
Length	min	Minimum duration
	max	Maximum duration
	points	Score
Posting-Time	start	Hashtag name
	end	Popularity
	points	Score
	engagement	Level of engagement

Methodology

Methodology



Results

Results

```
=====
CONTENT VIRAL POTENTIAL ANALYZER
=====

Available Content Types:
1. Comedy
2. Memes
3. Music
4. Dance
5. Pets
6. Education
7. Gaming
8. Technology
9. Beauty
10. Fitness
11. Food
12. Science
13. Art
14. Photography
15. Animation
16. Lifestyle
17. Travel
18. Fashion
19. DIY
20. Productivity
21. Finance
22. Business
23. History
24. Motivation
25. Automotive
26. Books
27. News
28. ASMR

Select Content Type (number): 11

Enter hashtags for your content: fyp, viral,trending,foodie,cooking
```

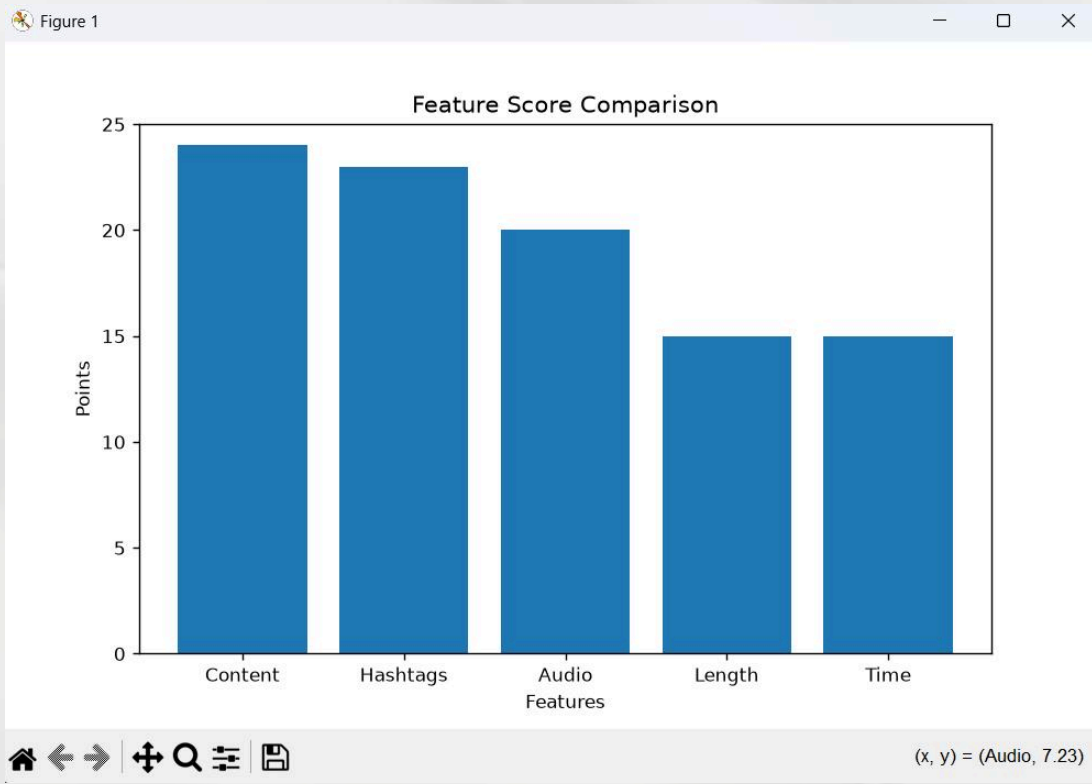
```
Available Audio Tracks:
1. Sigma Beat
2. Future Bass Drop
3. Dreamcore Loop
4. Speed Up Night
5. Hyper Pop Mix
6. Summer Anthem
7. Lofi Study
8. Soft Piano
9. Happy Pop
10. Night Drive
11. Chill Guitar
12. Acoustic Morning
13. Epic Cinematic
14. Jazz Mood
15. Relax Beats
16. Nature Sounds
17. Rain Ambience
18. Ocean Waves
19. Old Remix
20. Vintage Piano

Select Audio (number): 3

Video Length(seconds): 20

Posting hour (0-23):13

Analyzing...
```



```
=====
ANALYSIS RESULT
=====

Prediction : VIRAL
Total Score : 97/100
-----
Feature Scores
-----
Content Type : 24
Hashtags : 23
Audio : 20
Video Length : 15
Posting Time : 15
-----
Automata State History
-----
START
CONTENT_ANALYSIS
HASHTAG_ANALYSIS
AUDIO_ANALYSIS
QUALITY_ANALYSIS
SCORING
CLASSIFICATION
VIRAL
RECOMMENDATION
END
-----
Insights
-----
✓ The selected content type has good engagement potential.
✓ 2 hashtag(s) closely match the selected content.
✓ 3 general hashtag(s) detected.
✓ The selected audio has a good trend popularity.
✓ The video length is within the recommended range.
✓ The selected posting time falls within a peak engagement period.
-----
Recommendations
-----
No recommendations. Great job!
```


Results

```
CONTENT VIRAL POTENTIAL ANALYZER
=====

Available Content Types:
1. Comedy
2. Memes
3. Music
4. Dance
5. Pets
6. Education
7. Gaming
8. Technology
9. Beauty
10. Fitness
11. Food
12. Science
13. Art
14. Photography
15. Animation
16. Lifestyle
17. Travel
18. Fashion
19. DIY
20. Productivity
21. Finance
22. Business
23. History
24. Motivation
25. Automotive
26. Books
27. News
28. ASMR

Select Content Type (number): 8

Enter hashtags for your content: python, ai, fyp
```

```
Select Content Type (number): 8

Enter hashtags for your content: python, ai, fyp

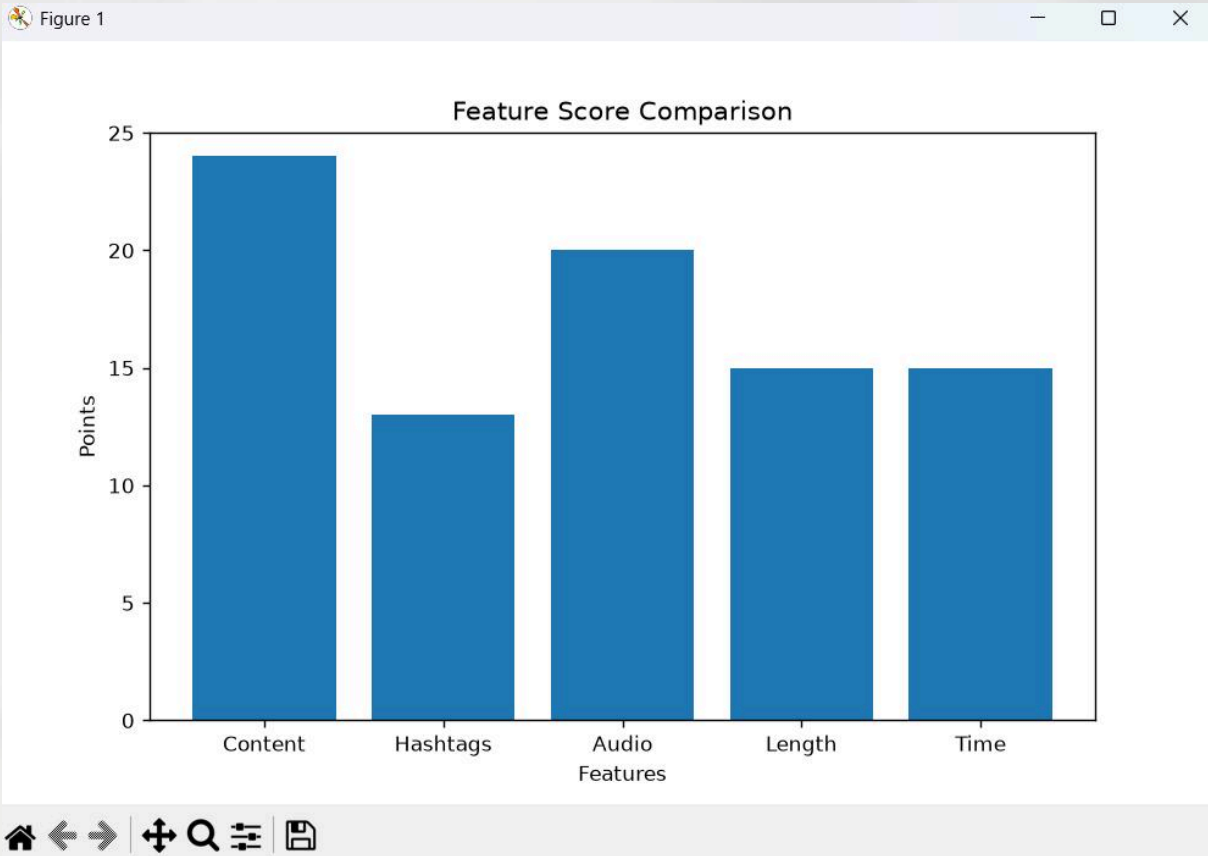
Available Audio Tracks:
1. Sigma Beat
2. Future Bass Drop
3. Dreamcore Loop
4. Speed Up Night
5. Hyper Pop Mix
6. Summer Anthem
7. Lofi Study
8. Soft Piano
9. Happy Pop
10. Night Drive
11. Chill Guitar
12. Acoustic Morning
13. Epic Cinematic
14. Jazz Mood
15. Relax Beats
16. Nature Sounds
17. Rain Ambience
18. Ocean Waves
19. Old Remix
20. Vintage Piano

Select Audio (number): 2

Video Length(seconds): 20

Posting hour (0-23):18

Analyzing...
```



```
=====
ANALYSIS RESULT
=====

Prediction   : HIGH
Total Score  : 87/100
-----

Feature Scores
-----

Content Type : 24
Hashtags     : 13
Audio        : 20
Video Length : 15
Posting Time : 15
-----

Automata State History
-----

START
CONTENT_ANALYSIS
HASHTAG_ANALYSIS
AUDIO_ANALYSIS
QUALITY_ANALYSIS
SCORING
CLASSIFICATION
HIGH
RECOMMENDATION
END
-----

Insights
-----

✓ The selected content type has good engagement potential.
✓ 2 hashtag(s) closely match the selected content.
✓ 1 general hashtag(s) detected.
✓ The selected audio has a good trend popularity.
✓ The video length is within the recommended range.
✓ The selected posting time falls within a peak engagement period.
-----

Recommendations
-----

No recommendations. Great job!
```


Results

```
=====
CONTENT VIRAL POTENTIAL ANALYZER
=====

Available Content Types:
1. Comedy
2. Memes
3. Music
4. Dance
5. Pets
6. Education
7. Gaming
8. Technology
9. Beauty
10. Fitness
11. Food
12. Science
13. Art
14. Photography
15. Animation
16. Lifestyle
17. Travel
18. Fashion
19. DIY
20. Productivity
21. Finance
22. Business
23. History
24. Motivation
25. Automotive
26. Books
27. News
28. ASMR

Select Content Type (number): 6

Enter hashtags for your content: books,study
```

```
Available Audio Tracks:
1. Sigma Beat
2. Future Bass Drop
3. Dreamcore Loop
4. Speed Up Night
5. Hyper Pop Mix
6. Summer Anthem
7. Lofi Study
8. Soft Piano
9. Happy Pop
10. Night Drive
11. Chill Guitar
12. Acoustic Morning
13. Epic Cinematic
14. Jazz Mood
15. Relax Beats
16. Nature Sounds
17. Rain Ambience
18. Ocean Waves
19. Old Remix
20. Vintage Piano

Select Audio (number): 7

Video Length(seconds): 60

Posting hour (0-23):10

Analyzing...
```

Recommendations

- Consider using hashtags more closely related to your content.

Suggestions:

- #studytok
- #learnontiktok
- #education

Reason: Some hashtags are less relevant to the selected content category.

- Consider adjusting the video length.

Suggestions:

- 15-30 seconds

Reason: Videos between 15 and 30 seconds receive the highest retention score.

- Consider posting during a peak engagement period.

Suggestions:

- 12:00-14:00
- 17:00-22:00

Reason: Peak posting times receive higher engagement scores in this project.

Figure 1

Feature Score Comparison

Feature	Score
Content	24
Hashtags	0
Audio	15
Length	9
Time	10

Points

(x, y) = (Audio, 2.66)

```
=====
ANALYSIS RESULT
=====

Prediction : MEDIUM
Total Score : 58/100

-----
Feature Scores
-----

Content Type : 24
Hashtags : 0
Audio : 15
Video Length : 9
Posting Time : 10

-----
Automata State History
-----

START
CONTENT_ANALYSIS
HASHTAG_ANALYSIS
AUDIO_ANALYSIS
QUALITY_ANALYSIS
SCORING
CLASSIFICATION
MEDIUM
RECOMMENDATION
END

-----
Insights
-----

✓ The selected content type has good engagement potential.
✓ 1 hashtag(s) are not found in the dataset.
✓ The selected audio has a good trend popularity.
```


Results

```
=====
CONTENT VIRAL POTENTIAL ANALYZER
=====

Available Content Types:
1. Comedy
2. Memes
3. Music
4. Dance
5. Pets
6. Education
7. Gaming
8. Technology
9. Beauty
10. Fitness
11. Food
12. Science
13. Art
14. Photography
15. Animation
16. Lifestyle
17. Travel
18. Fashion
19. DIY
20. Productivity
21. Finance
22. Business
23. History
24. Motivation
25. Automotive
26. Books
27. News
28. ASMR

Select Content Type (number): 27

Enter hashtags for your content: gaming, random
```

```
Available Audio Tracks:
1. Sigma Beat
2. Future Bass Drop
3. Dreamcore Loop
4. Speed Up Night
5. Hyper Pop Mix
6. Summer Anthem
7. Lofi Study
8. Soft Piano
9. Happy Pop
10. Night Drive
11. Chill Guitar
12. Acoustic Morning
13. Epic Cinematic
14. Jazz Mood
15. Relax Beats
16. Nature Sounds
17. Rain Ambience
18. Ocean Waves
19. Old Remix
20. Vintage Piano

Select Audio (number): 17

Video Length(seconds): 120

Posting hour (0-23):2

Analyzing...
```

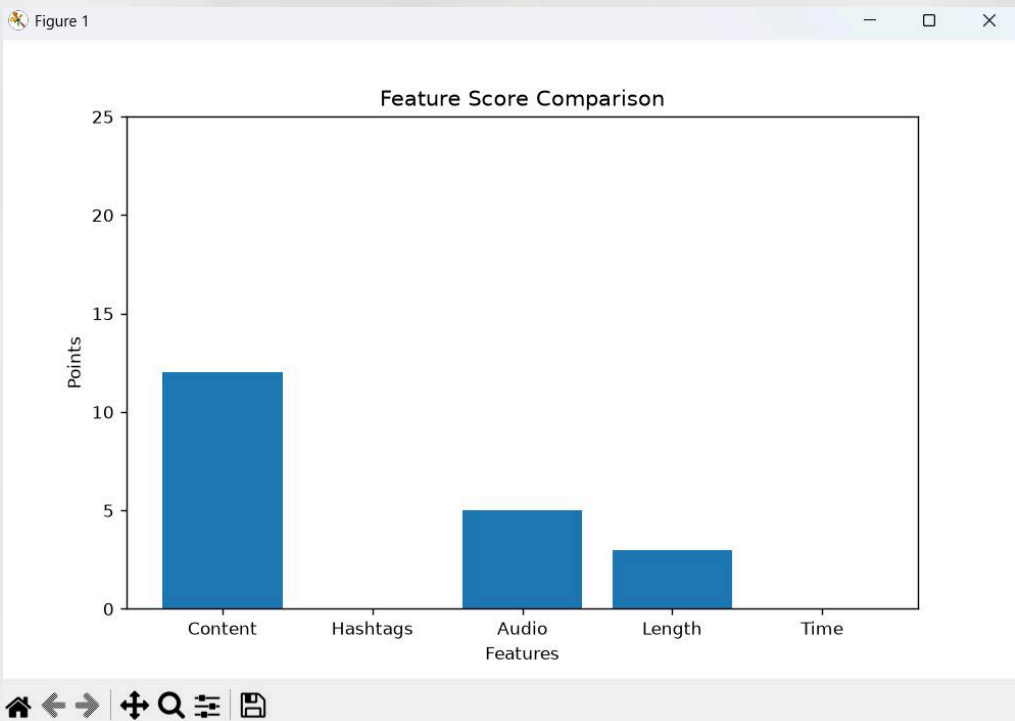
```
-----
Recommendations
-----

• Consider using hashtags more closely related to your content.
Suggestions:
- #news
Reason: Some hashtags are less relevant to the selected content category.

• Consider using a trending audio.
Suggestions:
- Sigma Beat
- Future Bass Drop
- Dreamcore Loop
Reason: Trending audio receives a higher score in the evaluation model.

• Consider adjusting the video length.
Suggestions:
- 15-30 seconds
Reason: Videos between 15 and 30 seconds receive the highest retention score.

• Consider posting during a peak engagement period.
Suggestions:
- 12:00-14:00
- 17:00-22:00
Reason: Peak posting times receive higher engagement scores in this project.
```



```
=====
ANALYSIS RESULT
=====

Prediction   : LOW
Total Score  : 20/100
-----

Feature Scores
-----

Content Type : 12
Hashtags     : 0
Audio        : 5
Video Length : 3
Posting Time  : 0
-----

Automata State History
-----

START
CONTENT_ANALYSIS
HASHTAG_ANALYSIS
AUDIO_ANALYSIS
QUALITY_ANALYSIS
SCORING
CLASSIFICATION
LOW
RECOMMENDATION
END
-----

Insights
-----

✓ The selected content type has moderate engagement potential.
✓ 1 hashtag(s) are not found in the dataset.
✓ The selected audio has moderate trend popularity.
```


Automata Design

IDT

Automata Design



Conclusion

Conclusion

- Built a data-driven content analysis system
- Used CSV datasets for scoring and evaluation
- Applied finite automata for classification
- Generated recommendations to improve content quality

audio.csv

data

1 audio_name,trend_level,points,genre,recommended_for

2 Sigma Beat,viral,20,Phonk,Gaming

3 Future Bass Drop,viral,20,EDM,Dance

4 Dreamcore Loop,viral,20,Ambient,Lifestyle

5 Speed Up Night,viral,20,Electronic,Sports

6 Hyper Pop Mix,viral,20,Pop,Dance

7 Summer Anthem,trending,15,Pop,Travel

8 Lo-fi Study,trending,15,Lo-fi,Education

9 Soft Piano,trending,15,Classical,Education

10 Happy Pop,trending,15,Pop,Lifestyle

11 Night Drive,trending,15,Synthwave,Automotive

12 Chill Guitar,trending,15,Acoustic,Travel

13 Acoustic Morning,average,10,Acoustic,Lifestyle

14 Epic Cinematic,average,10,Soundtrack,Technology

15 Jazz Mood,average,10,Jazz,Food

hashtags.csv

data

1 hashtag,trend_level,points,category,recommended_for

2 #fyp,viral,5,General,All

3 #viral,viral,5,General,All

4 #foryou,viral,5,General,All

5 #foryoupage,viral,5,General,All

6 #trending,viral,5,General,All

7 #explore,viral,5,General,All

8 #xyzbca,viral,5,General,All

9 #studytok,trending,4,Education,Education

10 #learnontiktok,trending,4,Education,Education

11 #education,trending,4,Education,Education

12 #python,trending,4,Technology,Technology

13 #datascience,trending,4,Technology,Technology

14 #machinelearning,trending,4,Technology,Technology

15 #ai,trending,4,Technology,Technology

content_types.csv

data

1 content_type,points,category,description

2 Comedy,30,Comedy,Humorous and entertaining videos

3 Memes,30,Memes,Trending internet humor

4 Music,30,Music,Singing or instrument performances

5 Dance,30,Dance,Dance challenges and choreography

6 Pets,30,Pets,Cute pets and animal videos

7 Education,24,Education,Tutorials and educational content

8 Gaming,24,Gaming,Gameplay and gaming highlights

9 Technology,24,Technology,Gadget reviews and tech news

10 Beauty,24,Beauty,Makeup and skincare content

11 Fitness,24,Fitness,Workout and fitness advice

12 Food,24,Food,Cooking and food reviews

13 Science,24,Science,Science facts and experiments

14 Art,24,Art,"Painting, drawing and creative work"

15 Photography,24,Photography,Photography and editing tips

length_rules.csv

data

1 min,max,rating,points,description

2 15,30,5,15,Excellent viewer retention

3 31,45,4,12,Very good engagement

4 46,60,3,9,Average engagement

5 61,90,2,6,Long video

6 91,600,1,3,Very long video

time_rules.csv

data

1 start,end,engagement,points

2 0,6,very_low,0

3 6,9,medium,7

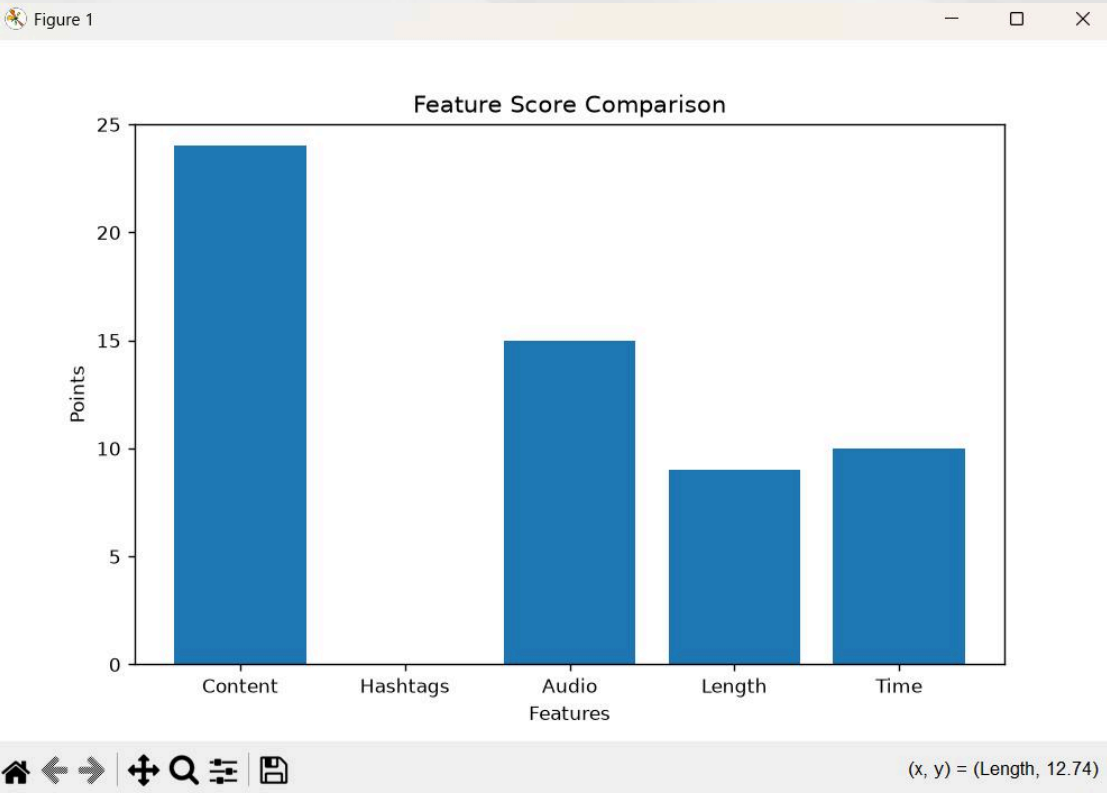
4 9,12,high,10

5 12,14,peak,15

6 14,17,high,10

7 17,22,peak,15

8 22,24,low,3



```
=====
ANALYSIS RESULT
=====

Prediction : MEDIUM
Total Score : 68/100

-----

Feature Scores
-----

Content Type : 18
Hashtags : 0
Audio : 20
Video Length : 15
Posting Time : 15
```

```
-----
Automata State History
-----

START
CONTENT_ANALYSIS
HASHTAG_ANALYSIS
AUDIO_ANALYSIS
QUALITY_ANALYSIS
SCORING
CLASSIFICATION
MEDIUM
RECOMMENDATION
END

-----

Insights
-----

✓ The selected content type has moderate engagement potential
✓ 4 hashtag(s) are not found in the dataset
✓ The selected audio is on trend
✓ The video length is within the recommended range
✓ The selected posting time is in peak engagement period
```


VOTE FOR ME!

**“VIRAL POTENTIAL
ANALYZER”**

IDT